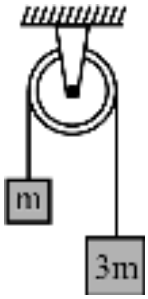
PHYSICS

1) If the system is released, then the acceleration of the block of mass m is (assume pulley and string is ideal) :-



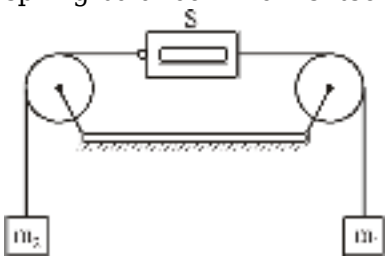
(1) g/4

(2) g/2

(3) g

(4) 2g

2) In the arrangement shown, the pulleys are fixed and ideal, the strings are light, $m_1 > m_2$ and S is a spring balance which is itself massless. The reading of S (in units of mass) is



(1) $m_1 - m_2$

(2) $\frac{1}{2}(m_1 + m_2)$

(3) $\frac{m_1 m_2}{m_1 + m_2}$

(4) $\frac{2m_1 m_2}{m_1 + m_2}$

3) A rocket of mass 6000 kg is set for vertical firing. If the exhaust speed be 1 km/s, how much gas must be ejected to give the rocket an upward acceleration of 20 ms^{-2}?(Take g = 10 ms^{-2})

(1) 45 kg/s

(2) 90 kg/s

(3) 120 kg/s

(4) 180 kg/s

4) Compute the least acceleration (in m/s^2) with which a 50 kg man can slide down a rope, if the rope can withstand a tension of 300 N.

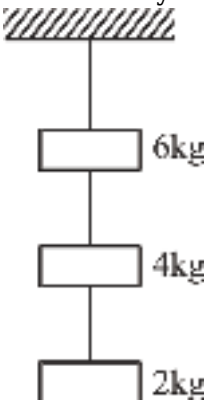
(1) 6

(2) 4

(3) 3

(4) 5

5) Three masses of 6kg, 4 kg and 2 kg are attached to a rigid support as shown in figure. If the string attached to the support breaks and the system falls freely, then the tension in the string connecting 4 kg and 2 kg mass is :



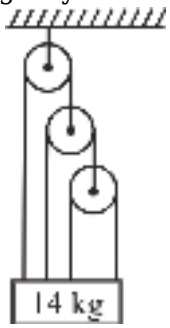
(1) Zero

(2) 8 kg wt

(3) 12 kg wt

(4) 6 kg wt

6) A 14kg block is hanged using a system of pulleys as shown in figure. Tension in string connecting celling and topmost pulley is :-



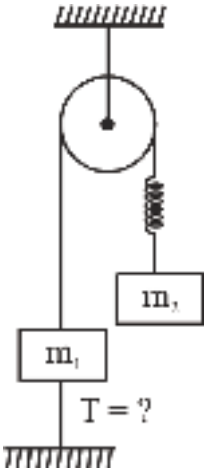
(1) 17.5 N

(2) 70 N

(3) 140 N

(4) 280 N

7) Mass m_2 is hanged using a uniform spring whose other end is attached to a string which is



attached to m_1. A string connects m_1 to ground as shown. Tension in this string is :-

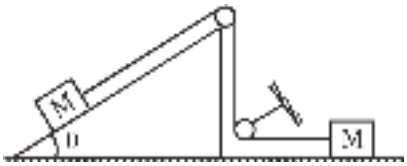
(1) zero

(2) m_1g

(3) m_2g

(4) $(m_2 - m_1)g$

8) Two blocks, each having a mass M, rest on frictionless surface as shown in the figure. If the pulleys are light and frictionless, and M on the incline is allowed to move down, then the tension in the string will be :-



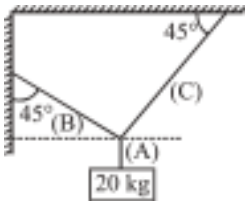
(1) 2.3 Mg sin θ

(2) 1.3 Mg sin θ

(3) $\frac{Mg}{2}\sin\theta$

(4) 2 Mg sin θ

9) In the given diagram, the tension in string C is:-

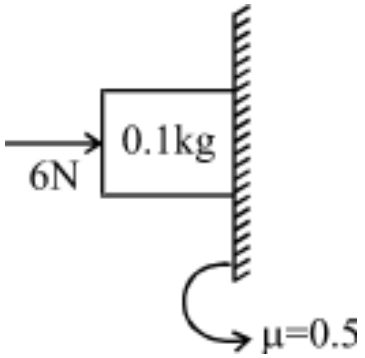


(1) 100 N

(2) 70.7 N

(3) 141 N

(4) 200 N



10) Find out frictional force acting on the block ? (g = 9.8 m/sec^2)

(1) 9.8 N

(2) 3 N

(3) 0.98 N

(4) 0.3 N

11) A block of mass 2 kg is kept on the floor. The coefficient of static friction is 0.4. If a force F of 2.5 N is applied on the block as shown in the figure, the frictional force between the block and the floor will be 

(1) 2.5 N

(2) 5 N

(3) 7.84 N

(4) 10 N

12) A car is moving on a straight horizontal road with a speed v. If the coefficient of friction between the tyres and the road us μ, distance in which the car can be stopped is :

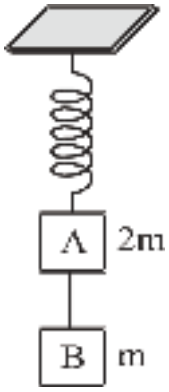
(1) $\frac{v^2}{2\mu g}$

(2) $\frac{v^2}{\mu g}$

(3) $\left(\frac{v}{\mu g}\right)^2$

(4) $\frac{v^2}{\mu}$

13) Two blocks A and B of masses 2m and m, respectively, are connected by a massless and inextensible string. The whole system is suspended by a massless spring as shown in the figure. The magnitude of acceleration of A and B, immediately after the string is cut, are respectively :



(1) g, g/2

(2) g/2, g

(3) g, g

(4) g/2, g/2

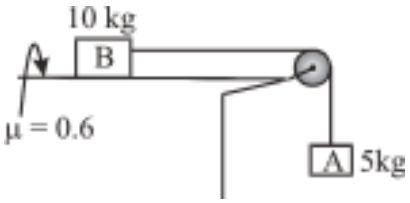
14) In the given arrangement, find the acceleration of mass 4kg at the instant when acceleration of block of mass 2kg is $2m/s^2$. 

(1) $4m/s^2$

(2) $2m/s^2$

(3) $5m/s^2$

(4) $6m/s^2$



15) Find the force of friction acting on block B. (Take g = 10 m/s^2) :-

(1) 36 N

(2) 60 N

(3) 50 N

(4) 600 N

16) A uniform chain of length L is resting on a table with a part of its hanging over the edge. If the co-efficient of static friction between the chain and the table is μ, the maximum length of the hanging part is :

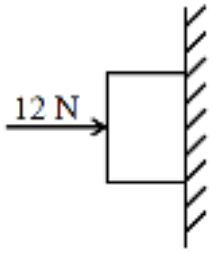
(1) $\frac{\mu L}{1+\mu}$

(2) $\frac{\mu L}{1-\mu}$

(3) $\frac{(1-\mu)L}{1+\mu}$

(4) μL

17) A block of weight 5N is pushed against a vertical wall by a force 12N. The coefficient of friction between the wall and block is 0.6. Find the magnitude of the force exerted by the wall on the block.



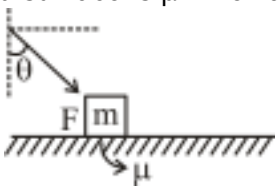
(1) 12 N

(2) 5 N

(3) 13 N

(4) 7.2 N

18) A block placed on a horizontal surface is being pushed by F force making an angle θ with vertical. The coefficient of friction between block and surface is μ. The value of force F required to slide the block with uniform velocity on the floor is -



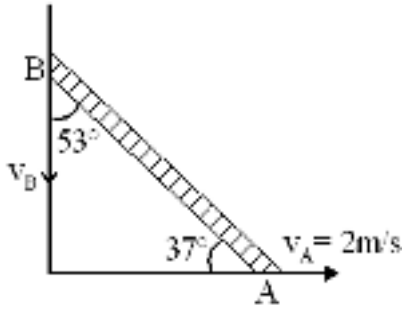
(1) $\frac{\mu mg}{\sin\theta - \mu\cos\theta}$

(2) $\frac{\sin\theta - \mu\cos\theta}{\mu mg}$

(3) μmg

(4) None

19) A ladder (AB) is placed against a wall on a floor at 37° from horizontal. If end 'A' is pulled horizontally with speed 2m/s, then find out downward velocity of end B.



(1) 6 m/s

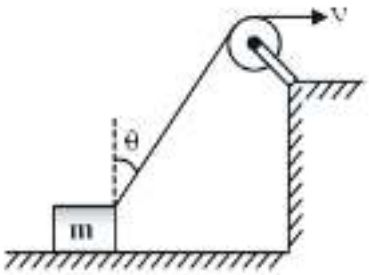
(2) 2 m/s

(3) $\frac{8}{3}$ m/s

(4) None of these

20) A block is dragged on a smooth plane with the help of a rope which moves with a velocity v as

shown in figure. The horizontal velocity of the block is :



(1) v

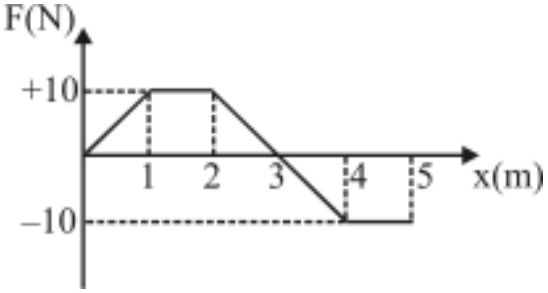
(2) $\frac{v}{\sin\theta}$

(3) v sin θ

(4) $\frac{v}{\cos\theta}$

21)

A position dependent force F acting on a particle and its force-position curve is shown in the figure. Work done on the particle, when its displacement 0 to 5 m is :-



(1) 35J

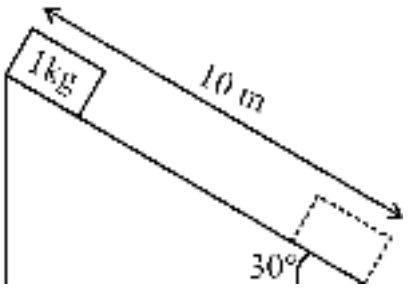
(2) 25J

(3) 15J

(4) 5J

22)

Find work done by gravitational force if block starts slipping down from top most point and reaches to the lower end:- $\left(g = 10\ m/s^2\right)$



(1) 50 J

(2) $-50\sqrt{3}$ J

(3) –100 J

(4) –200 J

23) The work done by a force $\vec{F} = \left(\hat{i} + 2\hat{j} + 3\hat{k}\right) N,$ to displace a body from position A to position B is :

[The position vector of A is $\vec{r}_1 = (\hat{i} + 3\hat{j} + \hat{k})m$ and position vector of B is $\vec{r}_2 = (2\hat{i} + 2\hat{j} + 3\hat{k})m$] -

(1) 5 J

(2) 3 J

(3) 2 J

(4) 10 J

24) A body of mass m is moving in a circle of radius r with a constant speed v. The force on the body is $\frac{mv^2}{r}$ and is directed towards the centre. What is the work done by this force in moving the body over half the circumference of the circle -

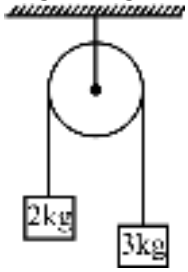
(1) $\frac{mv^2}{\pi r^2}$

(2) Zero

(3) $\frac{mv^2}{r^2}$

(4) $\frac{\pi r^2}{mv^2}$

25) In the figure shown find work done by tension in string on 3 kg block in first two second, from beginning of motion. The system is set free at t = 0 ($g = 10\ m/s^2$) (Assume pulley and string is ideal)



(1) –96 Joule

(2) –48 Joule

(3) +24 Joule

(4) 96 Joule

26) A particle is shifted from point (0, 0, 1) m to (1, 1, 2) m, under simultaneous action of several forces. Two of the forces are $\vec{F}_1 = (2\hat{i} + 3\hat{j} - \hat{k})N$ & $\vec{F}_2 = (\hat{i} - 2\hat{j} + 2\hat{k})N$. Find work done by resultant of these two forces.

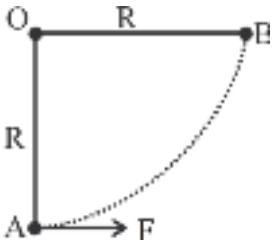
(1) 5J

(2) 3J

(3) 7J

(4) 9J

27) A block of mass m is taken from A to B under the action of a constant force F. Work done by this

force is :- 

(1) $\frac{\pi FR}{2}$

(2) πFR

(3) FR

(4) Zero

28) A force $F = Kx^2$ acts on a particle at an angle of 60° with the x–axis. the work done in displacing the particle from x_1 to x_2 will be –

(1) $\frac{kx^2}{2}$

(2) $\frac{k}{2}\left(x_2^2 - x_1^2\right)$

(3) $\frac{k}{6}\left(x_2^3 - x_1^3\right)$

(4) $\frac{k}{3}\left(x_2^3 - x_1^3\right)$

29) An experiment measures quantities a, b and c, and X is calculated from $X = \frac{ab^2}{c^3}$. If the percentage error in a, b and c are 1%, 3% and 2% respectively, the percentage error in X will be –

(1) 13%

(2) 7%

(3) 4%

(4) 1%

30) If v = ax + bt is dimensionally correct equation where (v is speed, x is distance and t is time) then dimensional formula of a and b are :-

(1) $[M^0 L^{-1} T^0]$, $[M^0 L^0 T^{-1}]$

(2) $[M^0 L^0 T^{-1}]$, $[M^0 L^1 T^{-2}]$

(3) $[M^0 L^0 T^{-1}]$, $[M^0 L^1 T^0]$

(4) $[M^0 L^{-2} T^{-1}]$, $[M^0 L^1 T^{-2}]$

31) If energy (E), velocity (V) and time (T) were chosen as fundamental physical quantities for measurement, then the dimensional formula for mass will be:-

(1) $[E^1 V^2 T^1]$

(2) $[E^2 V^{-2} T^0]$

(3) $[E^1 V^{-2} T^0]$

(4) $[E^{-1} V^2 T^1]$

32) The determination of the value of acceleration due to gravity)g) by simple pendulum method employs the formula, $g = 4\pi^2 \frac{L}{T^2}$. The expression for the relative error in the value of 'g' is :

(1) $\frac{\Delta g}{g} = 4\pi^2 \left[\frac{\Delta L}{L} - 2\frac{\Delta T}{T}\right]$

(2) $\frac{\Delta g}{g} = 4\pi^2 \left[\frac{\Delta L}{L} + 2\frac{\Delta T}{T}\right]$

(3) $\frac{\Delta g}{g} = \frac{\Delta L}{L} - 2\frac{\Delta T}{T}$

(4) $\frac{\Delta g}{g} = \frac{\Delta L}{L} + 2\frac{\Delta T}{T}$

33) One main scale division of a vernier callipers is a cm and n^{th} division of the vernier scale coincide with $(n - 1)^{th}$ division of the main scale. The least count of the callipers (in mm) is

(1) $\frac{10na}{(n-1)}$

(2) $\frac{10a}{(n-1)}$

(3) $\left(\frac{n-1}{10n}\right) a$

(4) $\frac{10a}{n}$

34) The least count of a stop watch is 0.2 second. The time of 20 oscillations of a pendulum is measured to be 25 seconds. The percentage error in the time period is

(1) 16%

(2) 0.8%

(3) 1.8%

(4) 8%

35) The length of a rectangular plate is measured by a meter scale and is found to be 10.0 cm. Its width is measured by vernier callipers as 1.00 cm. The least count of the meter scale and vernier callipers are 0.1 cm and 0.01 cm respectively (Obviously). Maximum permissible error in area measurement is -

(1) + 0.2 cm^2

(2) + 0.1 cm^2

(3) + 0.3 cm^2

(4) Zero

36) The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are:-

(1) 4, 4, 2

(2) 5, 1, 2

(3) 5, 1, 5

(4) 5, 5, 2

37) If A,B and C are physical quantities, having different dimensions, which of the following combinations can never be a meaningful quantity ?

(i) $\frac{AB}{C}$

(ii) $\frac{A+B}{C}$

(iii) $AB-C^2$

(iv) (A–B)C

(1) (iii, iv)

(2) (i, ii)

(3) (ii, iv)

(4) (ii, iii)

38) The circular scale of a screw gauge has 100 divisions and the pitch is 2mm. Find the true value of thickness of a thin sheet :-



(1) 2.16 mm

(2) 4.32 mm

(3) 6.16 mm

(4) None

39)

If velocity V, time period T and force F are used as fundamental physical quantities, then the dimensional formula of work is

(1) [VTF]

(2) $[V^{-1}T^{-1}F]$

(3) $[V^{-1}TF]$

(4) $[VT^{-1}F]$

40) The density of mercury is 15600 kg-m^{-3}. Its value in CGS system will be :-

(1) 15.6 g-cm^{-3}

(2) 1560 g-cm^{-3}

(3) 156 g-cm^{-3}

(4) 1.56 g-cm^{-3}

41) The resistance is $R = \frac{V}{I}$ where V = 100 ± 5 volts and I = 10 ± 0.2 amperes. What is the total error in R ?

(1) 5 %

(2) 7 %

(3) 5.2 %

(4) 0 %

42) The equation of the stationary wave is :

$$y = 2A\sin\left(\frac{2\pi ct}{\lambda}\right)\cos\left(\frac{2\pi x}{\lambda}\right).$$

Which statement is not true?

(1) The unit of ct is same as that of λ

(2) The unit of x is same as that of λ

(3) The unit of 2πc/λ is same as that of 2πx/λt

(4) The unit of c/λ is same as that of x/λ

43) Measure of two quantities along with the precision of respective measuring instrument is :-
A = 2.5 m/s ± 0.5 m/s B = 0.10s ± 0.01 s
The value of AB will be :-

(1) (0.25 ± 0.08)m

(2) (0.25 ± 0.5) m

(3) (0.25 ± 0.05) m

(4) (0.25 ± 0.135) m

44) Two resistances are given as $R_1 = (10 \pm 0.5)\Omega$ and $R_2 = (15 \pm 0.5)\Omega$. The percentage error in the measurement of equivalent resistance when they are connected in parallel is

(1) 6.33

(2) 2.33

(3) 4.33

(4) 5.33

45) The diameter of a cylinder is measured using a Vernier callipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is :-

(1) 5.112 cm

(2) 5.124 cm

(3) 5.136 cm

(4) 5.148 cm

CHEMISTRY

1) K_p for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ at 400ºC is 1.64×10^{-4}, then K_c for the above reaction is :-

(1) 0.5

(2) 0.93

(3) 1.27

(4) 3.89

2) For $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$ reaction started only with $NH_4HS(s)$ the observed pressure for reaction mixture in equilibrium is 1.8 atm at 120°C. What is the value of K_p for reaction :-

(1) 0.9 atm^2

(2) 8.1 atm^2

(3) 0.81 atm^2

(4) 1.8 atm^2

3) $AB_3(g)$ dissociates as

$AB_3(g) \rightleftharpoons AB_2(g) + \frac{1}{2} B_2(g)$.

When the initial pressure of AB_3 is 800 torr and the total pressure developed at equilibrium is 900 torr. What fraction of $AB_3(g)$ is dissociated ?

(1) 10%

(2) 20%

(3) 25%

(4) 30%

4) If low pressure and low temperature are the favourable conditions for the reaction:

$aA + bB \rightleftharpoons cC + dD$

then the true statements will be:

(1) (a + b) < (c + d) and $\Delta H = +x$

(2) (a + b) > (c + d) and $\Delta H = +x$

(3) (a + b) < (c + d) and $\Delta H = -x$

(4) no relation between ΔH and K_{eq}

5)

8 moles of a gas AB_3 are introduced into a 1.0 dm^3 vessel. It dissociates as

$2AB_{3(g)} \rightleftharpoons A_{2(g)} + 3B_{2(g)}$

At equilibrium 2 mole of A_2 are found to be present. The equilibrium constant of the reaction is -

(1) 2 $mol^2 L^{-2}$

(2) 3 $mol^2 L^{-2}$

(3) 27 $mol^2 L^{-2}$

(4) 36 $mol^2 L^{-2}$

6) For the hypothetical reactions, the equilibrium constant (K) values are given :
$A \rightleftharpoons B$; $K_1 = 2$
$B \rightleftharpoons C$; $K_2 = 4$
$C \rightleftharpoons D$; $K_3 = 3$
the equilibrium constant (K) for the reaction $D \rightleftharpoons A$ is -

(1) 24

(2) $\frac{1}{24}$

(3) 48

(4) 6

7) The value of K_c for the reaction;
$2A \rightleftharpoons B + C$ is 1×10^{-3}.
At a given time, the composition of reaction mixture is [A] = [B] = [C] = 1×10^{-4} M. In which direction the reaction will proceed?

(1) forward

(2) backward

(3) at equilibrium

(4) None of the above

8) The equilibrium constant for the reaction $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ is 25. If the volume of the container is reduced to half of the original volume the value of equilibrium constant will be

(1) 5

(2) 50

(3) 25

(4) 75

9) **Assertion (A):-** At equilibrium, concentration of reactants and products are always equal
Reason (R):- At equilibrium, rate of forward reaction and backward reaction is same

(1) Both A and R are true and the R is the correct explanation of the A.

(2) Both A and R are true but the R is not the correct explanation of the A.

(3) A is true but R is false.

(4) A is false R is true.

10) At a certain temperature, equilibrium constant (K_c) is 16 for the reaction :
$SO_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$
If we take one mole of each of the four gases in one litre container, what would be the equilibrium concentration of NO and NO_2 respectively?

(1) 0.4 mole, 1.6 mole

(2) 1.6 mole, 0.4 mole

(3) 0.6 mole, 0.4 mole

(4) 0.4 mole, 0.6 mole

11) The equilibrium constant for the given reaction :

$SO_3(g) \rightleftharpoons SO_2(g) + \frac{1}{2}O_2(g)$; $K_c = 4 \times 10^{-2}$

The value of K_c for the reaction :

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, will be

(1) 625

(2) 25

(3) 1/625

(4) None of these

12) $AgNO_3(s) \rightleftharpoons Ag(s) + NO_2(g) + \frac{1}{2}O_2(g)$

If total pressure at equilibrium is P and only $AgNO_3$ was taken initially then K_p ?

(1) $\frac{2}{3^{1/2}} P^{3/2}$

(2) $\frac{2}{3^{1/2}} P^{1/2}$

(3) $\frac{2}{3^{3/2}} P^{3/2}$

(4) None of these

13) The gas absorbs 100 J heat and is simultaneously compressed by a constant external pressure of 1.50 atm from 8 lit to 2 lit volume. Hence ΔU will be :

(1) – 812 J

(2) 812 J

(3) 1011 J

(4) 911 J

14) Predict which of the following reaction (s) has a negative entropy change ?

I. $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(\square)$

II. $NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$

III. $2KClO_4(s) \rightarrow 2KClO_3(s) + O_2(g)$

(1) III

(2) II

(3) I and II

(4) I

15) A system undergoes a process in which ΔE = + 300J while absorbing 400 J of heat energy and undergoing an expansion against 0.5 atm. What is the change in the volume (in L) ?

(1) 4

(2) 5

(3) 2

(4) 3

16) 2 mole of an ideal gas at 27°C expands isothermally and reversibly from a volume of 4 litre to 40 litre. The work done (in kJ) by the gas is :

(1) w = –28.72 kJ

(2) w = –11.488 kJ

(3) w = –5.736 kJ

(4) w = –4.988 kJ

17) Which of the following is/are state function :-

(1) E

(2) H

(3) S

(4) All of these

18) The entropy change involved in isothermal reversible expansion of 2 moles of an ideal gas from a volume of $10dm^3$ to a volume of $100dm^3$ at 27ºC is :-

(1) 42.3 J $mol^{-1}K^{-1}$

(2) 38.3 J $mol^{-1}K^{-1}$

(3) 35.8 J $mol^{-1}K^{-1}$

(4) 32.3 J $mol^{-1}K^{-1}$

19) Predict the sign of heat & work if heat is absorbed by the system & work is done by the system :-

(1) +q, –w

(2) –q, –w

(3) +q, +w

(4) –q, + w

20) For a certain reaction at 27°C the change in enthalpy and change in entropy are 40.63 kJ mol^{-1} and 100 JK^{-1} mol^{-1} respectively. The reaction is:-

(1) Spontaneous

(2) Non spontaneous

(3) Data is insufficient

(4) None

21) For a chemical reaction
$A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + 2D_{(g)} + xCal$:-

(1) Not possible at any temperature

(2) Possible only at low temperature

(3) Possible at all temperature

(4) Possible only at high temperature

22) For the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$; ΔH = ?

(1) $\Delta U - RT$

(2) $\Delta U - 2RT$

(3) $\Delta U + RT$

(4) $\Delta U + 2RT$

23) One mole of an ideal gas at 25°C expands in volume from 1.0 L to 4.0 L at constant temperature. What work (in J) is done if the gas expands against vacuum (Pexternal = 0) ?

(1) -4.0×10^2

(2) -3.0×10^2

(3) -1.0×10^2

(4) Zero

24) The work done by a system is 8J when 40J heat is supplied to it. The change in internal energy of the system during the process :-

(1) 32 J

(2) 40 J

(3) 48 J

(4) – 32 J

25)

For a gaseous reaction,
$A(g) + 3B(g) \rightarrow 3C(g) + 3D(g)$
ΔE is 17 Kcal at 27^0C assuming R=2 cal K^{-1} mol^{-1}, the value of ΔH for the above reaction is :-

(1) 15.8 Kcal

(2) 18.2 Kcal

(3) 20.0 Kcal

(4) 16.4 Kcal

26) An ideal gas undergoing isothermal expansion in vacuum shows :-

(1) $\Delta U = 0$

(2) $W = 0$

(3) $q = 0$

(4) All of these

27) In which one of the sets, all properties belong to the same category (all extensive or all intensive) ?

(1) Mass, Volume, Pressure

(2) Temperature, Pressure, Volume

(3) Heat capacity, Density, Mass

(4) Enthalpy, Internal energy, Volume

28) The correct thermodynamic conditions for the spontaneous reaction at all temperatures is

(1) $\Delta H < 0$ and $\Delta S = 0$

(2) $\Delta H > 0$ and $\Delta S < 0$

(3) $\Delta H < 0$ and $\Delta S > 0$

(4) $\Delta H < 0$ and $\Delta S < 0$

29) Which of the following hybridisation is termed as diagonal hybridisation?

(1) sp

(2) sp^2

(3) sp^3

(4) sp^3d

30) Hybridisation of Silicon in SiO_2 is :-

(1) sp

(2) sp^2

(3) sp^3

(4) sp^3d

31) Which of the following molecules are planar ?

(1) NH_3

(2) BrF_3

(3) SF_4

(4) CH_4

32) The correct geometry and hybridisation for XeF_4 are :-

(1) Tetrahedral, sp^3

(2) Square Planar, sp^3d^2

(3) Octahedral, sp^3d^2

(4) See-Saw, sp^3d

33) Which of the following molecules cannot have coordinate bond?

(1) $N_2H_5^{\oplus}$

(2) NO_2

(3) $Br_3^{\ominus}$

(4) CO_3^{-2}

34) Which of the following represents incorrect bond angle order ?

(1) $CH_4 > NH_3 > H_2O$

(2) $BCl_3 < AlCl_3 < GaCl_3$

(3) $PF_3 < PCl_3 < PBr_3$

(4) $NO_2^{\oplus} > NO_2 > NO_2^{\ominus}$

35) Which of the following molecule has zero dipole moment?

(1) BrF_3

(2) BF_3

(3) XeF_6

(4) SeO_2

36) Which of the following contains both polar and non-polar bonds?

(1) HCN

(2) H_2O_2

(3) CH_4

(4) CO_2

37) Which of the following represents correct order of dipole moment?

(1) $H_2O > H_2S$

(2) $NH_3 > PH_3$

(3) $H_2O > HF$

(4) All of these

38) Which one of the following species does not exist?

(1) Li_2

(2) F_2

(3) Be_2

(4) B_2

39) Which of the following has the minimum bond length?

(1) $O_2^{\oplus}$

(2) $O_2^{\ominus}$

(3) O_2^{-2}

(4) O_2

40) Which of the following diatomic molecular species has only pi bonds according to Molecular Orbital Theory?

(1) O_2

(2) N_2

(3) C_2

(4) Be_2

41) Resonance is not shown by :-

(1) C_6H_6

(2) CO_2

(3) SO_3

(4) SiO_2

42) The correct order of increasing C–O bond length is-

(1) $CO_3^{-2} < CO_2 < CO$

(2) $CO_2 < CO_3^{-2} < CO$

(3) $CO < CO_3^{-2} < CO_2$

(4) $CO < CO_2 < CO_3^{-2}$

43) Which of the following is correct order?

(1) $H_2O > NH_3 > NF_3 > \overset{\oplus}{C}H_3$ (dipole moment)

(2) $BF_3 = BCl_3 < AlCl_3$ (bond angle)

(3) $Cl_2 > Br_2 > F_2 < I_2$ (Bond dissociation energy)

(4) CO_3^{-2} > CO (C–O bond strength)

44) If 2s-2p mixing is ignored then which of the following will change its magnetic behaviour :-

(1) B_2

(2) C_2

(3) Both 1 and 2

(4) None

45) The formal charges on the three O-atoms in O_3 molecule are :-

(1) 0, 0, 0

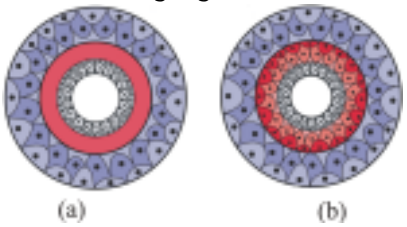
(2) 0, 0, –1

(3) 0, 0, +1

(4) 0, +1, –1

BIOLOGY

1) Following figures (a) and (b) represented the condition animals which are respectively :-



(1) (a)-Diploblastic, (b)-Diploblastic

(2) (a)-Triploblastic, (b)-Triploblastic

(3) (a)-Diploblastic, (b)-Triploblastic

(4) (a)-Triploblastic, (b)-Diploblastic

2) Consider the following statements regarding symmetry and coelom :
I. All triploblastic animals possess a true coelom.
II. Radial symmetry is characteristic of both Cnidarians and adult Echinoderms.
III. Pseudocoelomates are bilaterally symmetrical.
Which of the statements are correct?

(1) I and II only

(2) II and III only

(3) I and III only

(4) I, II, and III

3) Which of the following is a distinguishing feature of Osteichthyes (bony fish) compared to Chondrichthyes (cartilaginous fish)?

(1) Presence of placoid scales.

(2) Absence of an operculum.

(3) Possess a swim bladder (or air bladder).

(4) Are exclusively marine.

4) Complete the given correlation –
Air bladder : _____ :: Tailed amphibian : _____.

(1) *Exocoetus, Hyla*

(2) *Clarias, Salamandra*

(3) Dog fish, *Salamandra*

(4) Saw fish, *Ichthyophis*

5) Read the following statements and choose the correct option.
(A) In Mollusca, the space between the hump and the mantle is called the mantle cavity.

(B) In Echinodermates, digestive system is complete with mouth on the lower (ventral) side.
(C) Hemichordata was earlier considered as a class under phylum Chordata.
(D) In Mollusca, A hard and spongy layer of skin forms a mantle over the visceral hump.

(1) A-T, B-T, C-F, D-F

(2) A-F, B-T, C-T, D-F

(3) A-T, B-F, C-T, D-T

(4) A-F, B-T, C-F, D-T

6) Match the columns:-

Column-I		**Column-II**	
(a)	Pheretima	(i)	Lac insect
(b)	Laccifer	(ii)	Sea lily
(c)	Aplysia	(iii)	Earthworm
(d)	Antedon	(iv)	Sea hare

(1) a-iii; b-i, c-ii; d-iv

(2) a-i, b-iii; c-ii; d-iv

(3) a-i; b-iii; c-iv; d-ii

(4) a-iii; b-i; c-iv; d-ii

7) Read the following statements carefully and answer the question given below:
(a) Cyclostomes have circular and sucking mouth without jaw.
(b) Chondrichthyes have gills without operculum.
(c) Osteichthyes have 5-7 pairs of Gills covered by Operculum.
(d) In Amphibia, alimentary canal, urinary and reproductive tract open into Anus.
How many of the following options is/are incorrect?

(1) b and c

(2) c and d

(3) only d

(4) b and d

8) Match the following subphylum of phylum chordata with their characters and examples:

Subphylum	**Characters**	**Examples**
(A) Urochordata	(a) Notochord replaced by vertebral column	(i) Amphioxus
(B) Cephalochordata	(b) Notochord is present only in larval tail	(ii) Ascidia
(C) Vertebrata	(c) Notochord present from head to tail region and is persistent throughout their life	(iii) Fish

(1) A-b-ii ; B-c-i ; C-a-iii

(2) A-a-ii ; B-c-i ; C-b-ii

(3) A-c-i ; B-a-iii ; C-b-ii

(4) A-b-iii ; B-a-i ; C-c-ii

9) The teeth in class Chondrichthyes are modified _____ scales.

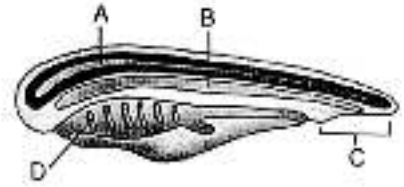
(1) Placoid

(2) Cycloid

(3) Ctenoid

(4) Ganoid

10) Given below is a diagrammatic representation of general chordate characteristics



Which of the following labelled structure is mesodermal in origin and is replaced by vertebral column in vertebrates?

(1) A

(2) B

(3) C

(4) D



11) Consider following diagram and select correct statement regarding this :-

(1) Triploblastic, Bilaterally symmetrical, Muscular pharynx

(2) 'a' is male, 'b' is female

(3) Dioecious, Pseudocoelomate, Internal fertilization.

(4) All are correct

12) In some animals, the body is externally and internally divided into segments with a serial wise repetition of at least some organs. Such phenomenon is known as :

(1) Dimorphism

(2) Metamerism

(3) Metagenesis

(4) Polymorphism

13) How many are amniotes & warm blooded ?
Chelone, *Hemidactylus*, *Struthio*, *Rohu*, *Salamander*, Human, Whale, *Psittacula*

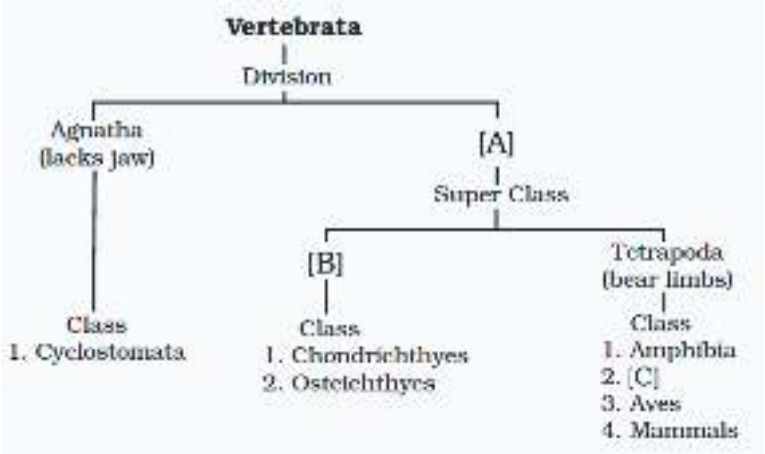
(1) 4

(2) 5

(3) 6

(4) 3

14) Find out the correct options for A,B & C :-



	[A]	[B]	[C]
(1)	Gnathostomata	Reptilia	Pisces
(2)	Reptilia	Pisces	Agnathostomata
(3)	Gnathostomata	Pisces	Reptilia
(4)	Pisces	Reptilia	Agnathostomata

(1) 1

(2) 2

(3) 3

(4) 4

15) Which of the following pair of characters are not true for Echinodermata, Hemichordata and Chordata ?

(1) Triploblastic, deuterostomia

(2) Deuterostomia, segmented

(3) Organ-system and triploblastic

(4) Organ-system, deuterostomia

16) Find out incorrect statement :-

(1) Phylum chordata includes animals which possess a notochord either throughout or during early embryonic life

(2) All of the vertebrates possess jaws.

(3) Amphibia, Reptilia, Aves and Mammalia have two pairs of limbs and are thus grouped under Tetrapoda.

(4) Fishes, amphibians and reptiles are poikilothermous.

17)

Read the following features with regards to Arthropoda :-
(a) They are triploblastic and bilateral symmetrical.
(b) Mostly oviparous
(c) Coelomate
(d) Circulatory system-closed type
(e) They are mostly dioecious

How many features are incorrect ?

(1) Two

(2) One

(3) Four

(4) Three

18)

You are observing an animal closely and you find some characters such as :
· Endoskeleton made up of cartilage.
· Mouth is located ventrally.
· The skin is tough, containing placoid scales.
· Notochord persist throughout life
· Air bladder is absent.

After observing the above features, in which group will you classify that animal :-

(1) Chondrichthyes

(2) Osteichthyes

(3) Urochordata

(4) Cephalochordata

19) Which of the following statements are correct or incorrect regarding class Amphibia :
(i) Body is divisible into head and trunk. Tail is present in some amphibians
(ii) Show respiration by gills, lungs and through, skin
(iii) Has scales in all its members
(iv) Can live dual life
(v) Has eyelids

(1) All are correct

(2) (i) and (iv) are correct

(3) Only (iii) is incorrect

(4) Only (ii) is incorrect

20) Which of the following statements are ture(T) and which are false(F) ? Choose the correct option :-

I. Amphibians have neurogenic heart.
II. The skull of mammals is dicondylic
III. In birds, jaws are modified into beak
IV. Voice is produced in Aves by a syrinx
V. Rabbit belongs to Reptilia.

(1) II, IV and V are true, I and III are false

(2) II, III and IV are true, I and V are false

(3) II and V are true, I, III and V are false

(4) I, II and V are true, III and IV are false

21) Read the following features :-
(i) Commonly known as 'round worm'.
(ii) Animals are bilateral symmterical and triploblastic.
(iii) Exclusively parasite on animals.
(iv) Pseudocoelomate animals having organ-system level of body organisation.
(v) Body shows segmentation.
(vi) Alimentary canal complete with well developed muscular pharynx.
(vii) Animals are bisexual and shows sexual dimorphism.
(viii) Fertilization internal and direct or indirect development.
Out of these how many features are not exhibited by members of Aschelminthes ?

(1) Two

(2) Three

(3) Four

(4) One

22)

Read the following four statements A –D :
(A) The classification does not helps in assigning a systematic position to a newly discovered species
(B) As over a million species of animals have been discovered till now, so the need of classification has become all the more important
(C) Notochord is a mesodermally derived rod-like structure formed on the dorsal side during embryonic development in all animals.
(D) In closed type circulatory system the blood is circulated through a series of blood vessels of same diameter (arteries, veins and capillaries)

How many of the above statements are incorrect ?

(1) Two

(2) Three

(3) Four

(4) One

23) **Assertion (A) :** Sponges are mostly asymmetrical.
Reason (R) : Any plane that passes through the centre does not divide in to equal halves.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

24) **Assertion :** Metamerism is a phenomenon that is exhibited by Annelida, Arthropoda and chordates.
Reason : Body is externally and internally divided into segments with a serial repetition of at least some organs.

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion

(2) Both Assertion and Reason are correct and Reason is not correct explanation of Assertion

(3) Assertion is true but Reason is false

(4) Both Assertion and Reason are false

25) During contraction of muscle :

(1) Thin filaments slides over the thick filaments.

(2) Thick filament slides over the thin filaments

(3) Both filaments slide over each other.

(4) No filament slides during contraction.

26) A joint between humerus and radio Ulna is :-

(1) Ball & socket

(2) Pivot

(3) Gliding

(4) Hinge

27) Which of the following type of joint allows considerable movements and therefore, play a significant role in locomotion?

(1) Synovial joints

(2) Fibrous joints

(3) Cartilagenous joints

(4) All of the above

28) Age related disorder characterised by decreased bone mass and increased chances of fractures is :

(1) Gout

(2) Arthritis

(3) Osteoporosis

(4) Both 1 and 2

29) In the alimentary canal and reproductive tract which type of muscles are present :

(1) Striated, involuntary

(2) Striated, voluntary

(3) Unstriated, voluntary

(4) Unstriated, involuntary

30) Vertebrochondral ribs of humans :-

(1) Are represented by 9th, 10th and 11th rib pairs

(2) Don't articulate directly with the sternum

(3) Join to the 7th and 11th rib

(4) Are represented by 11th and 12th rib pairs

31) Find out incorrect statement:

(1) All locomotions are movement, but all movements are not locomotions

(2) Red muscle fibres having more mitochondria and myoglobin

(3) Skeletal muscles are voluntary and striated

(4) Cardiac muscles are unstriated, branched and involuntary.

32)

Fill in the blanks
(i) ___a_____ is the only bone which is not attached with any other bone of the body.
(ii) Cavity of sphenoid bone is called ____b____ in which ____c____ is present.

(1)	a. Mandible	b. Sella turcica	c. Pituitary gland
(2)	a. Hyoid	b. Sella turcica	c. Hypothalamus
(3)	a. Hyoid	b. Sella turcica	c. Pituitary gland
(4)	a. Hyoid	b. Sella turcica	c. Pineal gland

(1) 1

(2) 2

(3) 3

(4) 4

33)

Read the following statements (A – D) :-
(A) A bands are dark and contain myosin and actin filaments
(B) I bands are light and cantain actin filaments
(C) Part between two Z-line is sarcomere
(D) Central part of A-band, not overlapped by thin filament is H-zone

How many of the above statements are correct ?

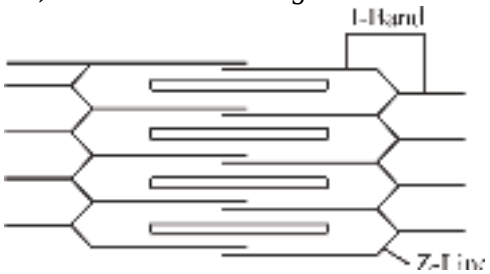
(1) Four

(2) One

(3) Two

(4) Three

34) Which of following statement is incorrect about given below diagram ?



(1) According the diagram maximally relaxation occur

(2) H-zone disappear clearly

(3) Distance between two successive Z-lines will be short

(4) Thin filament slides over the thick filament

35) Match the Column-I and Column-II and select the correct option ?

	Column I		**Column II**
(A)	Tetany	(i)	Paralysis of skeletal muscle
(B)	Muscular dystrophy	(ii)	Inflammation of joints
(C)	Arthritis	(iii)	Degeneration of skeletal muscles
(D)	Myasthenia gravis	(iv)	Rapid spasm in muscle

(1) B-i, A-ii, C-iii, D-iv

(2) A-iv, B-iii, C-ii, D-i

(3) D-iii, B-ii, C-i, A-iv

(4) A-iv, C-ii, B-i, D-iii

36) Rapid spasm in muscle is due to low concentration of _____ ion in body fluid

(1) Na^{+}

(2) Cl^{-}

(3) Ca^{2+}

(4) Mg^{2+}

37) Put the following statement in proper order to describe muscle contraction.
(1) Signal sent by CNS via motor neuron.
(2) Generation of action potential in the sarcolemma.
(3) Release of Ca^{+2} from sarcoplasmic reticulum.
(4) The neurotransmitter acetylcholine released at motor endplate.
(5) Sarcomere shorterns.

(1) 1 → 2 → 4 → 3 → 5

(2) 1 → 4 → 2 → 3 → 5

(3) 1 → 4 → 3 → 2 → 5

(4) 5 → 4 → 3 → 2 → 1

38)

How many of the following bones are unpaired?

Hyoid, Maxilla, Temporal, Mandible, Vomer, Sternum, Zygomatic

(1) Four

(2) Three

(3) Five

(4) Six

39) Match the column-I and II and select the correct option ?

Column-I			**Column-II**
A.	Fibrous Joint	(i)	Between tarsals
B.	Cartilaginous Joint	(ii)	Between the carpals
C.	Pivot Joint	(iii)	Between the cranial bones
D.	Hinge Joint	(iv)	Between the two adjoining vertebrae
E.	Gliding Joint	(v)	Between Atlas and Axis
		(vi)	Between Femur and Tibia
		(vii)	Between humerus and radioulna

Options :-

	A	**B**	**C**	**D**	**E**
1	(iii)	(iv)	(i), (ii)	(vi), (vii)	(v)
2	(iv)	(v)	(vi), (vii)	(iii)	(i), (ii)
3	(iii)	(iv)	(v)	(vi), (vii)	(i), (ii)
4	(iv)	(i), (ii)	(v)	(vi), (vii)	(iii)

(1) 1

(2) 2

(3) 3

(4) 4

40) Choose the correct matching :-

	Synovial Joint		**Bones Involved**
(a)	Ball and socket	(i)	Carpal and metacarpal of thumb
(b)	Hinge	(ii)	Atlas and axis
(c)	Pivot	(iii)	Frontal and parietal
(d)	Saddle	(iv)	Knee
		(v)	Humerus and pectoral girdle

(1) a-v, b-iv, c-ii, d-i
(2) a-ii, b-v, c-iv, d-i
(3) a-i, b-ii, c-v, d-iv
(4) a-v, b-iv, c-iii, d-i

41) The sporophyte of most of the mosses is composed of:

(1) Foot only
(2) Foot and capsule
(3) Foot, Seta and Capsule
(4) Capsule only

42) An aquatic fern is:

(1) Dryopteris
(2) Lycopodium
(3) Salvinia
(4) Equisetum

43) Evolutionary the first terrestrial plants to possess vascular tissue - Xylem and phloem is:

(1) Bryophytes
(2) Thallophytes
(3) Angiosperms
(4) Pteridophytes

44) Select the mismatched pair.

(1) *Pinus* - Dioecious
(2) *Selaginella* - Heterosporous
(3) *Funaria* - Multicellular rhizoids
(4) *Ulothrix* - Filamentous algae

45) The gymnosperms are heterosporous. They produce :

(1) All same types of spores
(2) Haploid microspores and megaspores
(3) Haploid microspores and diploid megaspores
(4) Diploid microspores and haploid megaspores

46) Which of the following shows diplontic life cycle?

(1) *Funaria*
(2) *Wolfia*
(3) *Azolla*
(4) *Albugo*

47) What is common in all three, Funaria, Dryopteris and *Pinus* ?

(1) Independent sporophyte
(2) Presence of archegonia
(3) Well developed vascular tissues
(4) Independent gametophyte

48) Life cycle pattern is Haplo-diplontic in :

(1) *Ustilago*
(2) *Ulothrix*
(3) *Fucus*
(4) *Funaria*



49) Identify the given figures :-

(1) *Salvinia*
(2) *Ulothrix*
(3) *Sphagnum*
(4) *Pinus*

50) Where does reduction division take place in the life cycle of liverwort ?

(1) Sporangium
(2) Capsule
(3) Ascus

(4) Ovule

51) How many egg cells could present in Archegonium of Bryophytes :-

(1) One
(2) Two
(3) Four
(4) Three

52) Which statements are true for bryophyta ?

(1) They have antheridium and Oogonium
(2) They have vascular tissue
(3) They bear roots
(4) Some of the members have leaf like structures

53) Which of the following is not a part of bryophyta?

(1) Liverwort
(2) Bryopsida
(3) True mosses
(4) Pteropsida

54) Antherozoids of bryophytes are

(1) Non-flagellated
(2) Cilliated
(3) Multi cilliated
(4) Biflagellated

55) Gemmae are structures of asexual reproduction among ..... ?

(1) Liver worts
(2) Mosses
(3) Pteridophyte
(4) Algae

56) Protonema is produced in :-

(1) *Riccia*
(2) *Funaria*
(3) *Marchantia*
(4) *Selaginella*

57) A protonema is :-

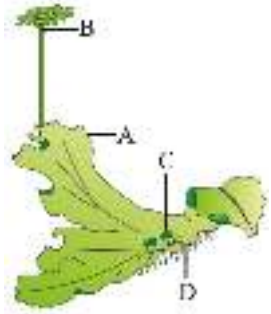
(1) A structure in pteridophytes formed before the thallus develops

(2) A sporophytic free living structure formed in pteridophytes

(3) A creeping, green, filamentous and gametophytic structure produced in bryophytes.

(4) A primitive structure formed after fertilization in pteridophytes

58) Examine the figure given below and choose the structure in which asexual buds are produced :-



(1) A

(2) B

(3) C

(4) D

59) Which of the following have multicellular sporophyte which is not free living but attached to photosynthetic gametophyte and derives nourishment from it ?

(1) Pteridophytes

(2) Gymnosperms

(3) Bryophytes

(4) Thallophytes

60) In mosses vegetative reproduction take place by:-

(1) fragmentation and budding in the secondary protonema

(2) gemmae formation and endospore formation

(3) gemmae and tuber formation

(4) Spores

61) Most reduced gametophyte present in :

(1) *Funaria*

(2) *Pteris*

(3) *Lilium*

(4) *Equisetum*

62) Which of the following provide peat as fuel and used as packing material for trans-shipment of living material ?

(1) *Sargassum*

(2) *Sphagnum*

(3) *Polytrichum*
(4) *Funaria*

63) Great ecological importance of mosses :-

(1) Packing material for tran-shipment
(2) Help in succession along with lichen
(3) Food for birds
(4) Peat fuel

64) Branched & unbranched stem are found respectively in :

(1) *Cedrus* & *Pinus*
(2) *Pinus* & *Cedrus*
(3) *Cycas* & *Pinus*
(4) *Cedrus* & *Cycas*

65) Match Column I with Column II.

	Column-I		**Column-II**
(a)	Sphenopsida	(i)	*Psilotum*
(b)	Lycopsida	(ii)	*Selaginella*
(c)	Pterosida	(iii)	*Equisetum*
(d)	Psilopsida	(iv)	*Pteris*

(1) a-i, b-ii, c-iv, d-iii
(2) a-ii, b-iii, c-i, d-iv
(3) a-iii, b-ii, c-iv, d-i
(4) a-ii, b-iii, c-iv, d-i

66) Which is incorrect :

(1) Asexual reproduction in liverwort take place by gemma
(2) Mosses have an elaborate mechanism of spore dispersal
(3) Pteridophytes are used for medicinal purpose and as a soil binders
(4) All pteridophytes are homosporous in nature

67) Which of the following is correct

(1) The stems are branched in Cycas
(2) Coralloid roots are associated with N_2-fixing fungi
(3) The gymnosperms are homosporous
(4) All spermatophytes are heterosporous in nature

68) In gymnosperms :-

(1) Male and female gametophytes are independent of sporophyte
(2) Male and female gametophytes are dependent on sporophyte
(3) Ovary represents megasporangium
(4) Embryo sac represents female gametophyte

69) In gymnosperms, pollination is carried out by

(1) Animals
(2) Birds
(3) Water
(4) Air currents

70) The leaves of gymnosperms are well adapted to withstand extremes of environmental conditions due to

(1) Being broad and hard
(2) Presence of thick cuticle and sunken stomata
(3) Presence of superficial stomata
(4) Presence of fruit

71) Endosperm in gymnosperms is :

(1) n
(2) 2n
(3) 3n
(4) 4n

72) *Adiantum* belongs to which of the given class of pteridophyte?

(1) Lycopsida
(2) Sphenopsida
(3) Psilopsida
(4) Pteropsida

73) Which among the following pteridophytes produces two kinds of spores?

(1) *Selaginella*
(2) *Equisetum*
(3) *Salvinia*
(4) Both (1) and (3)

74) Sporophylls which form a distinct compact structure called strobili or cones are found in :

(1) *Salvinia*
(2) *Funaria*
(3) *Selaginella*

(4) *Sphagnum*

75) In which of the given gymnosperms, coralloid roots are associated with N_2-fixing cyanobacteria?

(1) *Pinus*
(2) *Cedrus*
(3) *Ginkgo*
(4) *Cycas*

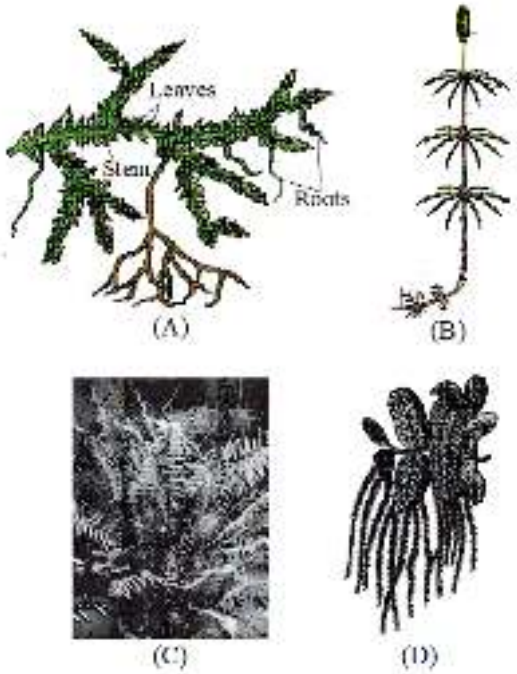
76) Select the **correct** statement

(1) The leaves in gymnosperms are not well-adapted to withstand extreme of temperature, humidity and wind
(2) Gymnosperms have naked seeds
(3) *Selaginella* belongs to class Psilopsida
(4) All pteridophytes are homosporous

77) The giant redwood tree is :

(1) *Pinus*
(2) *Cedrus*
(3) *Sequoia*
(4) *Cycas*

78) Go through the following figures and identify these plants (A,B,C and D) :-



(1) A-*Equisetum*, B-*Selaginella*, C-Fern, D-*Salvinia*
(2) A-*Selaginella*, B-*Equisetum*, C-Fern, D-*Salvinia*

(3) A-Fern, B-*Salvinia*,
C-*Equisetum*, D-*Selaginella*

(4) A-*Salvinia*, B-*Equisetum*,
C-Fern, D-*Selaginella*

79) Read the following paragraph and fill in the blanks with approprirte words/ terms.
In pteridophytes the main plant body is a ____(A)____which bears sporangia that are subtended by leaf like appendages called ___(B)____ The sporangia produce spores by ___(C)____ and after germination of spores, they give rise to photosynthetic gametophytes called ____(D)____.
Choose the correct option :-

	A	B	C	D
(1)	Sporophyte	sporangia	meiosis	Protonema
(2)	Gametophyte	Sporophylls	mitosis	Protonema
(3)	Sporophyte	Sporophylls	meiosis	Prothallus
(4)	Gametophyte	Sporophylls	mitosis	Prothallus

(1) 1

(2) 2

(3) 3

(4) 4

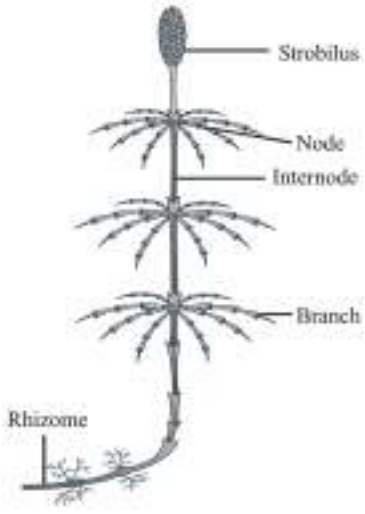
80) **Assertion (A)** :- In gymnosperms, the male and female gametophytes do not have an independent existence. **Reason (R)** :- They are dependent on sporophyte for growth and development.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is not correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true



81) Study the given diagram and give answer of questions a, b and c.
(a) Name of plant ? (b) Class of plant ?
(c) Ploidy of rhizome ?

	(a)	(b)	(c)
(1)	*Sphagnum*	*Lycopsida*	Haploid
(2)	*Funaria*	Moss	Haploid
(3)	*Funaria*	Liverwort	Diploid
(4)	*Equisetum*	*Sphenopsida*	Diploid

(1) 1

(2) 2

(3) 3

(4) 4

82) Arrange the following events in a correct sequence in the life cycle of fern :-
(I) Fertilization (II) Liberation of spores
(III) Prothallus (IV) Embryo

(1) II, III, I, IV

(2) IV, III, II, I

(3) I, II, III, IV

(4) I, IV, III, II

83) Which of the following statement is **correct** for pteridophyte ?

(1) Majority of pteridophytes are heterosporous

(2) Majority of pteridophytes are homosporous

(3) Majority of pteridophytes have seed

(4) Majority of pteridophytes are fruit

84) Identify the true statements for the plant shown in figure and select the correct option given



below :- (A) Monoecious gametophyte of this plant require cool, damp, shady place to grow
(B) Antheridia and archegonia both are formed on same gametophyte
(C) It produce two different type of spore.
(D) The development of the zygote into young embryo takes place within the female gametophyte.

(1) A, B, C & D

(2) B, C & D only

(3) C & D only

(4) A, C & D only

85) Answer the following questions (a, b, c and d) by using the given diagram :-



(a) Name of plant ?

(b) Ploidy of plant ?

(c) Plant is homosporus or heterosporus ?

(d) Which type of event is shown by this plant ?

	(a)	(b)	(c)	(d)
(1)	Selaginella	Diploid	Heterosporus	Precursor to seed habit
(2)	Salvinia	Diploid	Heterosporus	Precursor to seed habit
(3)	Pinus	Haploid	Homosporus	Fruit habit
(4)	Riccia	Haploid	Homosporus	Origin of seed

(1) 1

(2) 2

(3) 3

(4) 4

86) Match the column :-

Column-I		Column-II	
(a)	Jan Ingenhousz	(i)	First action spectrum of photosynthesis
(b)	Joseph priestley	(ii)	Essentiality of sunlight to plants
(c)	T. W. Engelmann	(iii)	Discovered oxygen

(1) a-(i), b-(ii), c-(iii)

(2) a-(ii), b-(i), c-(iii)

(3) a-(iii), b-(ii), c-(i)

(4) a-(ii), b-(iii), c-(i)

87) Which of the following is incorrect about the experiment performed by T.W. Engelmann ?

(1) *Cladophora* (a green algae) is used.

(2) Suspension of aerobic bacteria is used.

(3) Bacteria accumulated mainly in the region of orange and green light of split spectrum.

(4) Prism was used to split light into its spectral components.

88) $2H_2A + CO_2 \rightarrow 2A + CH_2O + H_2O$
If this reaction occurs in cyanobacteria then what is A :-

(1) S

(2) H_2O

(3) N_2

(4) O

89) Moll's half leaf experiment proves that :-

(1) Chlorophyll is essential for photosynthesis

(2) CO_2 is essential for photosynthesis

(3) O_2 released in photosynthesis comes from H_2O

(4) Both (1) and (2)

90) Photosynthesis is an :-

(1) Endergonic, oxidative catabolism.

(2) Exergonic, redox anabolism.

(3) Endergonic, redox anabolism.

(4) Exergonic, reductive catabolism.

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	4	2	1	3	4	3	3	3	1	1	2	3	3	1	3	1	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	1	2	1	1	3	3	1	2	3	4	4	2	1	2	3	2	1	1
Q.	41	42	43	44	45															
A.	2	4	1	3	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	3	3	3	3	2	2	3	4	2	1	3	3	3	3	2	4	2	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	2	4	1	2	4	4	3	1	3	2	2	4	2	2	2	4	3	1	3
Q.	86	87	88	89	90															
A.	4	4	1	3	4															

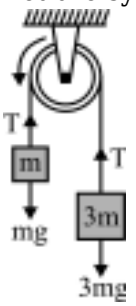
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	3	2	1	4	2	1	1	2	4	2	1	3	2	2	2	1	3	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	2	1	1	1	4	1	3	4	2	4	3	1	1	2	3	2	1	3	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	4	1	2	2	2	4	3	2	1	4	4	4	1	2	3	3	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	4	3	4	4	2	4	2	1	4	4	3	4	2	3	2	3	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	1	2	3	1	4	3	4	2	3										

SOLUTIONS

PHYSICS

1) **Explanation:** Two blocks are hanging with the help of a fixed pulley it is asked about the acceleration of the block of mass m
Concept: atwood machine, NLM
Solution:
Let the system moves with an acceleration a as shown in the figure.

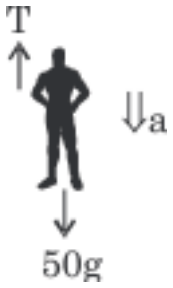


3mg – T = 3ma
T – mg = ma
2mg = 4ma
$a = \frac{g}{2}$
Acceleration of mass m is g/2 upward
$a = \frac{g}{2}$ upward
Final answer: correct option (2)

2) $m = \frac{T}{g} = \frac{2m_1m_2}{m_1 + m_2}$

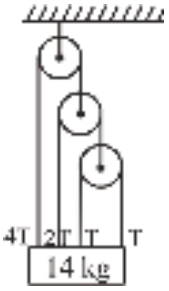
3) $a = \frac{\alpha V_{rel}}{M} - g \Rightarrow 20 = \frac{\alpha \times 1000}{6000} - 10$ $\Rightarrow \alpha = 180$ kg/s

4)



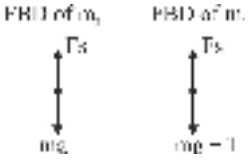
50g – T = 50a $a = 10 - \frac{300}{50} = 4$ m/s^2

5) In free fall, T = 0

6) 

Therefore, 4T + 2T + T + T = 14g

8T = 140 = Tension in string connecting ceiling and topmost pulley

7) 

For m_2 : Fs = m_2g

For m_1 : Fs = $m_1g + T$

$m_2g = m_1g + T$ ⇒ $T = (m_2 - m_1)g$

8) $a = \frac{Mg\sin\theta}{(M+M)} = \frac{g}{2}\sin\theta \therefore \quad T = Ma = \frac{Mg}{2}\sin\theta$

9)

Since $T_C \cos 45° = T_B \sin 45° \Rightarrow T_C = T_B$ ...(1)

$T_C \sin 45° + T_B \cos 45° = 20g \Rightarrow 2T_C \times \frac{1}{\sqrt{2}} = 200$

$T_C = 100\sqrt{2}$ N = 141 N

10) S.F. < f_L

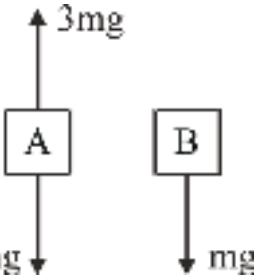
⇒ S.F. = f_L = 0.1 × 9.8 = 0.98 N

11) Applied force = 2.5 N and limiting friction = μmg = 0.4 × 2 × 9.8 = 7.84 N
As applied force is less than limiting friction. So, for the given condition static friction will work. Static friction on a body = Applied force = 2.5 N.

12) $0 = v^2 - 2as$

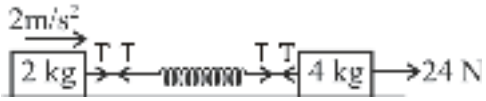
$\Rightarrow s = \frac{v^2}{2a}$

$= \frac{v^2}{2\mu g}$

13) 

$a_A = g/2$

$a_B = g$

14) 

T = 2(2) $\qquad$ 24 – T = 4 (a)

T = 4N $\qquad$ 24 – 4 = 4a

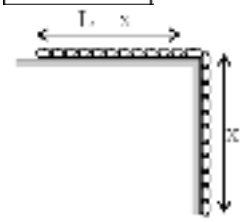
$\qquad\qquad$ a = 5 m/s^2

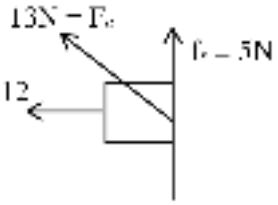
15) (fr)max = m(10)g = 60 N

but required force is 5g = 50 N

16) $\mu\left[\frac{M}{L}(L-x)g\right] = \frac{M}{L} \times xg$

$$\boxed{x = \frac{\mu L}{1+\mu}}$$





17)

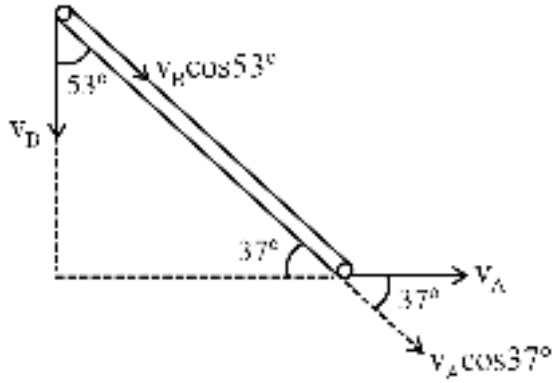
$$\boxed{f_\ell = 7.2N}$$

18) N = F cosθ + mg $\quad$ ...(1)

F sinθ = μN

F sinθ = μ(Fcosθ + mg)

$$F = \left(\frac{\mu mg}{\sin\theta - \mu\cos\theta}\right)$$



19)

$v_A\cos37° = v_B\cos53°$

$$v_A\left(\frac{4}{5}\right) = v_B\left(\frac{3}{5}\right)$$

$\Rightarrow 2 \times 4 = v_B(3)$

$$\Rightarrow v_B = \frac{8}{3} \text{ m/s}$$

20)

Question Explanation:

Find the horizontal velocity (v_b) of a block dragged by a rope moving at velocity v, where the rope makes an angle θ with the vertical.

Concept:

String Constraint — The component of the block's velocity along the rope must equal the rope's velocity v.

Solution:

Angle with horizontal = $(90^\circ - \theta)$.

Velocity component along rope: $v_b \cos(90^\circ - \theta) = v_b \sin\theta$.

Equation: $v_b \sin\theta = v \implies v_b = \frac{v}{\sin\theta}$.

Final Answer:

(Option 2)

21) **Explaining :** A position dependent force F acting on a particle and its force-position curve is shown in the figure. Work done on the particle, when its displacement 0 to 5 m is :-

Concept : This question is based on work with variable force

Solution : $W = (\text{Area})_1 - (\text{Area})_2$

$$W = \frac{1}{2} \times (3 + 1) \times 10 - \frac{1}{2} \times (2 + 1) \times 10$$

$$= 5 \text{ J}$$

Final Answer : (4) 5J

22) **Question Explanation :**

A 1 kg block slides down a 10 m inclined plane at 30°. We need to calculate the work done by gravity as it moves from the topmost point to the lower end.

Concept-Based (Work Done by Gravity on an Inclined Plane)

Formula Used :
$W = mg\sin\theta \cdot d$

Calculation :
Given

A. m = 1 kg,
B. g = 10 m/s^2,
C. θ = 30°,
D. d = 10 m.

$$W = (1 \times 10) \times \sin 30^\circ \times 10$$
$$W = 10 \times \frac{1}{2} \times 10 \quad W = 50\ J$$

Correct answer : (1)

23)

Explain Question : Calculate work done by constant force for given displacement.

Concept : Work done by constant force (dot product).

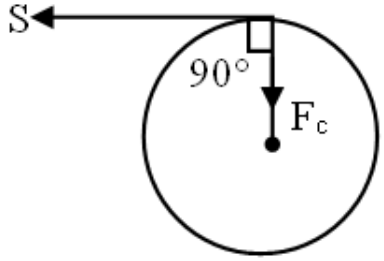
Solution :

$$\vec{s} = \vec{r}_2 - \vec{r}_1 = \hat{i} - \hat{j} + 2\hat{k}$$
$$W = \vec{F} \cdot \vec{s}$$
$$= (\hat{i} + 2\hat{j} + 3\hat{k}) \cdot (\hat{i} - \hat{j} + 2\hat{k})$$
$$= 5\ J$$

Final Answer : option (1)

24)

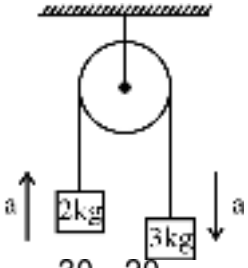
Concept
Work done by a force :
Formula ⇒
W = F·d = Fdcosθ
Since the angle between the centripetal force and the displacement is always 90°, we use:
cos90° = 0
which means the work done by the centripetal force is zero.
Mathematical Calculation
Since the angle between the force and displacement is 90°:



W = Fdcos90° = 0
Thus, for any motion along the circular path, the work done by the centripetal force is zero.
Final Answer
Correct Option: (2) Zero
Level : Easy

25)

Explain question :- we need to find the work done by the tension on the 3 kg block in first two second, from begining of motion.
Concept :- work done by tension force.
Solution :-



$a = \frac{30-20}{5} = 2 m/sec^2$

$T = \frac{2 \times 6 \times 10}{5} = 24$

displacement $= \frac{1}{2} \times 2 \times 2^2 = 4$

work done by tension on
3kg block = – T × d = –24 × 4 = –96 J
Final answer :- correct option (1)

26) $\vec{S} = (1-0)\hat{i} + (1-0)\hat{j} + (2-1)\hat{k}$

$= \hat{i} + \hat{j} + \hat{k}$

$\vec{F}_{Net} = (2+1)\hat{i} + (3-2)\hat{j} + (2-1)\hat{k}$

$\Rightarrow 3\hat{i} + \hat{j} + \hat{k}$

$W = \vec{F}.\vec{S} = (3 \times 1) + (1 \times 1) + (1 \times 1)$

$\Rightarrow$ 5J

27)

The component of displacement in the direction of force F is R.

 W = FR

28) $dW = kx^2 dx \cos 60°$

 $WD = \frac{k}{2}\int_{x_1}^{x_2} x^2 dx = \frac{k}{6}\left(x_2^3 - x_1^3\right)$

29) **Question Explanation : To find** percentage error in X where x is given by $X = \frac{ab^2}{c^3}$ and percentage error is given in a,b,c.
Concept : Error propagation
Solution :

$X = ab^2 / c^3 \Rightarrow \frac{\Delta X}{X} = \frac{\Delta a}{a} + 2\frac{\Delta b}{b} + 3\frac{\Delta c}{c}$

$\frac{\Delta X}{X} \times 100 = +1\% + 2 \times 3\% + 3 \times 2\% = 13\%$

Final Answer : option (1)

30)

V = ax + bt
so, [v] = [ax]

$$[a] = \left[\frac{v}{x}\right] = \left[\frac{LT^{-1}}{L}\right] = \left[T^{-1}\right]$$

for [b] = ?

[b] = [bt]

$$[b] = \left[\frac{v}{t}\right] = \frac{M^0L^1T^{-1}}{T} = \left[LT^{-2}\right]$$

31)

$M \propto E^x V^y T^z$
$M \propto [ML^2T^{-2}]^x [LT^{-1}]^y [T]^z$
$M^1L^0T^0 \propto M^x L^{2x+y} T^{-2x-y+z}$
On comparison
x = 1
2x + y = 0
2x + y = 0
y = –2

–2x – y + z = 0
–2 (1) – (–2) + z = 0
–2 + 2 + z = 0
z = 0

$$\boxed{M \propto E^1V^{-2}T^0}$$

32) $g = 4\pi^2\frac{L}{T^2}$

$$g \propto \frac{L}{T^2}$$

$$\boxed{\frac{\Delta g}{g} = \frac{\Delta L}{L} + \frac{2\Delta T}{T}}$$

33) $L.C = \left(\frac{b-a}{b}\right)^m$

$$L.C = \left(\frac{n-(n-1)}{n}\right) a \text{ cm}$$

$$= \frac{a}{n}\text{cm} = \frac{10a}{n}\text{mm}.$$

34) Δt = 0.2 s.
t = 25s

$$T = \frac{t}{N} \Rightarrow \frac{\Delta T}{T} = \frac{\Delta t}{t} = \frac{0.2}{25} = 0.8\%$$

35)

$A = \ell b = 10.0 \times 1.00 = 10.00$

$$\frac{\Delta A}{A} = \frac{\Delta \ell}{\ell} + \frac{\Delta b}{b}$$

$$\frac{\Delta A}{10.00} = \frac{0.1}{10.0} + \frac{0.01}{1.00}$$

$$\Rightarrow \Delta A = \left(\frac{1}{100} + \frac{1}{100}\right) = 10.00\left(\frac{2}{100}\right)$$

$= \pm 0.2\ cm^2$

36) $\underbrace{23.023}_{5SF}, \underbrace{0.0003}_{1SF}, \underbrace{2.1}_{2SF} \times 10^{-3}$

37)

Quantities having different dimensions can-not be added or subtracted.

38) Least count $= \frac{\text{Pitch}}{\text{No. of division}} = 0.02$ mm
Main scal reading = 2 (pitch) = 4 mm
Circular scale reading = 24
Zero error = (0.02 mm) 8 = 0.16 mm
True Value = Measured Value – Zero error
= 4.48 – 0.16
= 4.32 mm

39)

$W = V^aT^bF^c$
$ML^2T^{-2} = [LT^{-1}]^a[T]^b[MLT^{-2}]^c$
 a = 1, b = 1, c = 1
W = [VTF]

40) $n_2 = n_1\left[\frac{M_1}{M_2}\right]^a\left[\frac{L_1}{L_2}\right]^b\left[\frac{T_1}{T_2}\right]^c$
for density a = 1 b = – 3 c = 0

$$n_2 = 15600\left[\frac{kg}{gm}\right]^1\left[\frac{m}{cm}\right]^{-3}\left[\frac{s}{s}\right]^0$$

$= 15600 \times 1000 \times (100)^{-3}$
$n_2 = 15.6$

41) $R = \frac{V}{I}$

$\frac{\Delta R}{R} \times 100 = \left(\frac{\Delta V}{V} \times 100\right) + \left(\frac{\Delta I}{I} \times 100\right)$

$\frac{\Delta R}{R} \times 100 = \left(\frac{5}{100} \times 100\right) + \left(\frac{0.2}{10} \times 100\right)$

= 5 + 2

= 7 %

42) $\frac{C}{\lambda}$ = per sec

$\frac{x}{\lambda}$ = dimension less

43) **Question Explanation:** Find the result of multiplying two quantities along with the error.
Concept: Error Propagation in Multiplication.
Let y = AB

$\frac{\Delta y}{y} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$

$y = 2.5 \times 0.1 = 0.25$

$\frac{\Delta y}{0.25} = \frac{0.5}{2.5} + \frac{0.01}{0.10}$

$\Rightarrow \Delta y = 0.075 = 0.08$ (2SF)
Final Answer: 1

44) **Question Explanation:** Find the percentage error in the equivalent resistance of two parallel resistors with given resistances and errors.
Concept: Error propagation in parallel resistance
Solution: The equivalent resistance for two parallel resistors is $R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$. The percentage error is found by differentiating the reciprocal formula.

$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

Differentiating both sides, we get

$\frac{\Delta R}{R^2} = \frac{\Delta R_1}{\Delta R_1^2} + \frac{\Delta R_2}{R_2^2}$

$\left[R = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \times 15}{10 + 15} = 6\right]$

$\Rightarrow \frac{\Delta R}{R} = \left(\frac{\Delta R_1}{R_1^2} + \frac{\Delta R_2}{R_2^2}\right) R$

$= \left(\frac{0.5}{100} + \frac{0.5}{225}\right) 6$

$= \left(\frac{6 \times 0.5}{25}\right)\left(\frac{1}{4} + \frac{1}{9}\right) = \frac{13}{300}$

$\frac{\Delta R}{R} \times 100 = \frac{13}{3} = 4.33\%$

Final Answer: The percentage error in the measurement of equivalent resistance when they are connected in parallel is 4.33%.

45) 50 divisions = 2.45 cm

$\Rightarrow$ 1 division = $\frac{2.45}{50} = 0.049\text{cm}$

$\Rightarrow$ least count = 1 MSD –1VSD = 0.05 – 0.049
= 0.001 cm.

So vernier reading = 0.001 × 24 = 0.024 cm.

Therefore diameter of cylinder
= 5.10 + 0.024 = 5.124 cm.

CHEMISTRY

46)

Question Explanation: Convert K_p to K_c using relation involving Δn.

Concept: $K_p = K_c (RT)^{(\Delta n)}$
Δn = moles of gaseous products − reactants

Solution:
Δn = 2 − (1 + 3) = −2

T = 400°C = 673 K
R = 0.082 L atm mol^{-1} K^{-1}

$K_c = K_p (RT)^{(-\Delta n)} = K_p (RT)^2$

RT = 0.082 × 673 ≈ 55.2

$K_c = 1.64 \times 10^{-4} \times (55.2)^2$
$\approx 1.64 \times 10^{-4} \times 3047$
≈ 0.5

Final Answer: Option (1)

47)

$NH_4HS\ (s) \rightleftharpoons NH_3(g) + H_2S(g)$
0.9 0.9
$K_P = 0.9 \times 0.9 = 0.81$

48)

	AB_3	$\rightleftharpoons$	AB_2	+	$\frac{1}{2}B_2$
t=0	800		–		–
eq.	800-P		P		$\frac{P}{2}$

$800 - P + P + \frac{P}{2} = 900$

P = 200

$$\alpha = \frac{200}{800} \times 100\%$$

α = 25%

49)

Explaining : true statements will be:
Concept : This question is based on Le-Chatelier's principle
Solution :
Low P ⇒ V increases in forward direction
Hence, (c + d) > (a + b)
Low T ⇒ Reaction must be exothermic or ΔH < 0
Final Answer (3)

50) $2AB_{3(g)} \rightleftharpoons A_{2(g)} + 3B_{2(g)}$

8	-	-
8 – 2x	x	3x

Given, x = 2

$$K = \frac{2 \times (6)^3}{(4)^2} = 27$$

$27 mol^2 L^{-2}$

51)

Explaining : -
Concept : This question is based on
Solution :

$$A \rightleftharpoons D;\ K$$

$$K = K_1 \times K_2 \times K_3$$

$$\Rightarrow 2 \times 4 \times 3$$

$$\Rightarrow 24$$

$$D \rightleftharpoons A;\ K' = \frac{1}{K} = \frac{1}{24}$$

Final Answer : (2)

52)

Explaining : In which direction the reaction will proceed?
Concept : This question is based on reaction quotient.
Solution :

$$Q_C = \frac{[B][C]}{[A]^2} \Rightarrow \frac{1 \times 10^{-4} \times 1 \times 10^{-4}}{\left(1 \times 10^{-4}\right)^2} \Rightarrow 1$$

$Q_C > K_C$ so reaction will proceed in backward direction.
Final Answer : (2)

53) Equilibrium constant depends only on temeprature.

54) At equilibrium concentration of reactant and product becomes constant.

55) $So_2(g) + NO_2(g) \rightleftharpoons SO_3(g) + NO(g)$

Intial	1	1	1	1
cons.	$\frac{1-x}{1}$	$\frac{1-x}{1}$	$\frac{1+x}{1}$	$\frac{1+x}{1}$

$$K = \frac{[SO_3][NO]}{[SO_2][NO_2]} = \frac{(1+x)(1+x)}{(1-x)(1-x)}$$

$$16 = \frac{(1+x)^2}{(1-x)^2}, \frac{1+x}{1-x} = 4$$

$$x = \frac{3}{5} = 0.6$$

$[NO_2] = 1 - 0.6 = 0.4$

$[NO] = 1 + 0.6 = 01.6$

56) $$K_c = \left(\frac{1}{K}\right)^2$$

57) $AgNO_3(s) \rightleftharpoons Ag(s) + \underset{P_1}{NO_2(g)} + \frac{1}{2}\underset{\frac{P_1}{2}}{O_2(g)}$

$$\frac{3P_1}{2} = P$$

$$K_p = P_1\left(\frac{P_1}{2}\right)^{1/2}$$

$$\Rightarrow \frac{2P}{3}\left(\frac{2P}{2\times 3}\right)^{1/2} = \frac{2}{3^{3/2}}P^{3/2}$$

58)

$\Delta U = q + w$

$w = - P_{ext.}\Delta V$

59) If $\Delta ng < 0$ then entropy change is negative.

60)

Question Explanation: The system absorbs heat and expands against a constant external pressure. Given ΔE and q, find the volume change.

Concept: First law of thermodynamics:$\Delta E = q + w$
For expansion work: $w = -p_{ext}\,\Delta V$
1 atm·L = 101.325 J

Solution:

$\Delta E = +300$ J, $q = +400$ J
$w = \Delta E - q = 300 - 400 = -100$ J

$w = -\ p_{ext}\ \Delta V = -100$
$p_{ext}\ \Delta V = 100$

$\Delta V = 100 / (0.5 \times 101.325) \approx 1.97$ L ≈ 2 L

Final Answer: Option 3

61) $w = -nRT \ln \frac{V_2}{V_1}$

$w = -2 \times 8.314 \times 10^{-3} \times 300 \ln \frac{40}{4}$
$w = -11.488$ kJ

62)

Explain Question:
Asking about identifying state functions.

Concept:
State functions.

Solution:
State functions depend only on the current state, not path
Internal energy (E), enthalpy (H) and entropy (S) are all state functions

Final answer: Option (4)

63)

$\Delta S = 2.303\ nR \log_{10} \left(\frac{V_2}{V_1}\right)$

$= 2.303 \times 2 \times 8.314 \log \left(\frac{100}{10}\right) = 38.3$ J mol^{-1}K^{-1}

64)

Explanation:
Predict the signs of heat (g) and work (ω) when heat is absorbed by the system and work is done by the system.

Concept:
Sign Conventions in the First Law of Thermodynamics ($\Delta U = q + \omega$)

Solution:
According to standard thermodynamics convention for the First Law $\Delta U = q + \omega$:

• **Heat (q):** If heat is absorbed by the system q is positive (+ve).
• **Work (ω) :** If work is done by the system , |So, ω a negative (-ve).
Therefore q is positive and ω is negative.

Correct option:
(1) +q, –w

65) **Explain Question:**
Determine the spontaneity of a reaction at 27°C given the change in enthalpy (ΔH = +40.63kJ/mol) and the change in entropy (ΔS = +100J/(K mol)).

Concept:
Gibbs Free Energy and Spontaneity:ΔG = ΔH – TΔS

Solution:
ΔG = ΔH – TΔS
$\therefore \Delta G = 40.63 - \frac{300 \times 100}{1000} = 10.63 \text{ kJ mol}^{-1}$
therefore reaction will be non-spontaneous

Final answer: option (2)

66) **<u>Concept</u>:**
$\Delta H = -ve$ (exothermic reaction)
Δng > 0 (entropy increases so ΔS = +ve)
ΔG = ΔH – TΔs
$\Delta H = -ve, \Delta S = +ve \Rightarrow \Delta G = -ve$
(spontenous at all temperature)
<u>Correct Option</u>: (3)

67) **Explanation:**
Determine the relationship between enthalpy change (ΔH) and internal energy change (ΔU). for the reaction $N_2(g) + 3H_2(g) \leftrightharpoons 2NH_3(g)$.

Concept:
Relationship between Enthalpy Change (ΔH) and internal Energy Change (ΔU).

Solution:
The relationship is $\Delta H = \Delta U + \Delta n_g RT$
$\Delta n_g = (2) - (1 + 3) = 2 - 4 = -2$ mol.
So, ΔH = ΔU + (–2) RT
ΔH = ΔU – 2RT.

Correct option:
(2)

68) **Explain Question:** Asking about work done during expansion against vacuum
Concept: Free expansion work

Solution: Solution: For expansion against vacuum, external pressure $P_{ext} = 0$. Work $W = -P_{ext} \Delta V = 0 \times (4 - 1) = 0$ J.
Final answer: Option (4)

69) **Question Explanation**
Calculate the change in internal energy (ΔU) of a system when heat is supplied to it and work is done by the system.
Given Data
Work done by the system (W_{by}) = – 8J
Heat supplied to the system (q) = + 40 J
Concept
This question is based on First Law of Thermodynamics.
Mathematical Calculation
The First Law of Thermodynamics can be written as $\Delta U = q - W_{by}$
Substitute the values into the First Law:
ΔU = (+40 J) -(8J)
ΔU = (+40J) + (-8J)
ΔU = 32 J
Final Answer
The change in internal energy of the system is 32 J .
Option (1)

70)

Explain the Question : We are asked to find ΔH for a gaseous reaction given ΔE, using the relation between ΔH and ΔE for gases.
Concept : This question is based on for gaseous reactions : $\Delta H = \Delta E + \Delta ng\, RT$
Solution : Reaction : A(g) + 3B(g) → 3C(g) + 3D(g)
$\Delta ng = 6 - 4 = 2$
$\Delta n_g = RT = 2 \times (2 \text{ cal } K^{-1} \text{ mol}^{-1}) \times 300 \text{ K} = 1200 \text{ cal} = 1.2 \text{ kcal}$
$\Delta H = 17 + 1.2 = 18.2$ kcal
Final answer : 2

71)

For isothermal expansion in vacuum
ΔU = 0
W = 0
∵ ΔU = q + w
So, q = 0

72)

Explaining :-
In which of the given options all the properties are same (All extensive or all intensive)
concept :-
Intensive properties and extensive properties

solution :-
enthalpy, internal energy and volume are extensive properties.
Option (4)
Enthalpy internal energy , volume.

73) $\Delta G = \Delta H - T.\Delta S$
For, $\Delta H < 0$ and $\Delta S > 0$, ΔG = –ve (always)
 spontaneous at all temperatures.

74) NCERT Pg. # 121

75) NCERT Pg. # 329

76) NCERT Pg. # 115

77) NCERT Pg. # 115

78) NCERT Pg. # 110

79) NCERT Pg. # 108

80) NCERT Pg. # 111

81) NCERT Pg. # 111

82) NCERT Pg. # 112

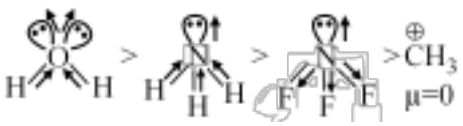
83) NCERT Pg. # 129

84) NCERT Pg. # 129

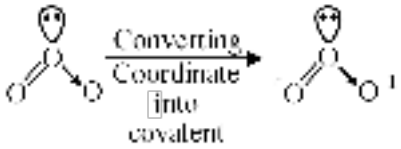
85) NCERT Pg. # 130

86) NCERT Pg. # 109

87) NCERT Pg. # 110

88) 

89) • When 2s-2p mixing applicable, then B_2 is paramagnetic and C_2 is diamagnetic.
• When 2s-2p mixing ignored, then
B_2 becomes diamagnetic.
C_2 becomes paramagnetic.

90) 

Therefore, the formal charges are 0, +1 and –1.

BIOLOGY

91) **Question Explanation:** Identifying animal germ layers.
Concept: Germ Layers
Solution: Figure (a) shows an animal with two germ layers, an outer ectoderm and an inner endoderm, separated by a non-cellular layer (mesoglea). This is a characteristic of diploblastic animals. Figure (b) shows an animal with three distinct germ layers: the outer ectoderm, the inner endoderm, and a middle mesoderm. This is a characteristic of triploblastic animals.
Final Answer: (a)-Diploblastic, (b)-Triploblastic

92) **Explaining :**
Concept :
Solution : I. Incorrect :
Platyhelminthes are triploblastic but are acoelomate (lack a coelom). Aschelminthes are triploblastic but pseudocoelomate.
II. Correct: Cnidarians (e.g., *Hydra*, jellyfish) and adult Echinoderms (e.g., starfish) both exhibit radial symmetry.
III. Correct: Pseudocoelomates (e.g., nematodes/roundworms) are bilaterally symmetrical.
Final Answer : (2)

93) **Explaining :**
Concept :
Solution : Bony fish (Osteichthyes) typically possess a swim bladder (or air bladder) for buoyancy regulation, which is generally absent in cartilaginous fish. Placoid scales and the absence of an operculum are characteristic of Chondrichthyes. Both groups have marine members.
Final Answer : (3)

94) **Explaining :**
Concept :
Solution : Dogfish and sawfish belong to Chondrichthyes, but air bladder is present in Osteichthyes. *Ichthyophis* is a limbless amphibian. Body of *Hyla* is divided into head and truck only.
Final Answer : (2)

95)

NCERT Page No. 45

96) NCERT Page No. 44,45

97) NCERT Page No.47

98) NCERT Page No. 56

99) NCERT Page No. 47

100) NCERT Page No. 46

101) NCERT XI Pg. # 52

102) NCERT XIth Pg.#48

103) NCERT Pg. # 59

104) NCERT Pg. # 56

105) NCERT (XIth) Pg. # 48

106) NCERT (XIth) Pg. # 61

107) NCERT Pg. # 53

108) NCERT (XI) Pg. # 56

109) **Question Explanation:** Identify which of the statements regarding class Amphibia are correct or incorrect.
Concept: Amphibia
Solution: The correct answer is option 3. Statement (iii) "Has scales in all its members" is incorrect. Amphibians are characterized by moist, scaleless skin. The other statements are correct.
Final Answer: 3

110)

Question Explanation

The question is asking which of the statements about vertebrate groups are true or false.

Concept
vertebrate characteristics

Solution
I. Amphibians have pronephric or mesonephric kidneys, not metanephric → False
II. Mammals have dicondylic skulls → True
III. Birds have beaks → True
IV. Birds produce voice by syrinx → True
V. Rabbit belongs to Mammalia, not Reptilia → False

Final answer
Option 2

111) NCERT XI (H) Pg. # 51, 52

112)

Question Explanation: : The question is asking how many of the given four statements about classification, notochord, and circulatory systems are incorrect.

Concept : This question is based on Basis of Classification.

Solution :
Statement A is **incorrect**. Classification does help in assigning a systematic position to newly discovered species.
Statement C is **incorrect**. The notochord is present only is some animals (chordates), not all animals.
Statement D is **incorrect**. In a closed circulatory system, blood flows through vessels, but arteries, veins, and capillaries differ in **diameter,** not same diameter.

Final Answer : option (2)

113)

Question Explanation: Identifying sponge symmetry.
Concept: Animal Body Symmetry
Solution: Both the **Assertion** and the **Reason** are true, and the Reason correctly explains the Assertion. The Assertion states that sponges are mostly **asymmetrical**, meaning they lack a plane of symmetry. The Reason correctly defines asymmetry as a condition where no plane passing through the center divides the body into equal halves. Therefore, the reason provides the direct definition for the asymmetrical condition mentioned in the assertion.
Final Answer: Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

114) **Question Explanation:**
This question is asking to evaluate the correctness of the given Assertion and Reason related to the concept of metamerism.

Concept :
Metamerism and Segmentation.

Explanation:
Assertion:
Metamerism is exhibited by Annelida, Arthropoda and Chordates.
This is correct. True metameric segmentation is a characteristic feature of these three groups.
Reason:
Metamerism means the body is externally and internally divided into segments with serial repetition of organs.
This is also correct because this definition accurately explains the concept of metamerism.
The reason correctly explains why these phyla show metamerism.

Final Answer: Option 1

115)

NCERT Pg.no 220

116)

Module

117)

NCERT Pg. # 228 Summary

118)

NCERT Pg. # 227

119)

NCERT Pg. # 218

120)

Question Explanation: Identify the correct statement about vertebrochondral ribs in humans.
Concept: Classification of human ribs

Solution: The correct answer is option 2

A. Human ribs are classified into:

    A. True ribs (1st–7th pairs) – Directly attached to sternum.

B. Vertebrochondral ribs (8th–10th pairs) – Indirectly attached to sternum through the cartilage of the 7th rib.

C. Floating ribs (11th–12th pairs) – Do not attach to sternum.

B. Vertebrochondral ribs do not directly articulate with the sternum, but attach indirectly via costal cartilage.

Final Answer: Option 2

121)

Question Explanation: The question asks to identify the incorrect statement about types of muscles and movement.

Concept: The question is based on locomotion and movement – muscle fibre types

Solution:

A. (1) All locomotions are movement, but all movements are not locomotions → Correct.
B. (2) Red muscle fibres having more mitochondria and myoglobin → Correct.
C. (3) Skeletal muscles are voluntary and striated → Correct.
D. (4) Cardiac muscles are unstriated, branched and involuntary → Incorrect. Cardiac muscles are **striated**, branched, and involuntary.

Final Answer: Option 4

122) **Explain Question:** Identify the correct bone and cavity details in the human skull.
Concept: Based on skull
Solution: The correct answer is Option 3.
• Hyoid(a) is the only bone not attached to any other bone.
• Sella turcica(b) is a cavity of the sphenoid bone.
• The pituitary gland(c) is present in the sella turcica.
Final answer: Option (3)

123) **Question Explanation**: The question asks how many statements about muscle fibre structure are correct.
Concept: The question is based on voluntary muscle.
Solution:The correct answer is **option 1 (Four)**.

A. **(A)** → Correct. A band is dark and contains both actin (thin) and myosin (thick) filaments.
B. **(B)** → Correct. I band is light and contains only actin filaments.
C. **(C)** → Correct. The region between two Z-lines is a sarcomere, the functional unit of muscle.
D. **(D)** → Correct. The central part of A-band not overlapped by actin is the H-zone.

All four statements are correct.

Final answer: Option 1

124)

NCERT - Pg. # 308

125) **Question Explanation :**
The question is asking to correctly match muscular and skeletal disorders with their descriptions.

Concept :
This question is based on muscular and skeletal disorders.

Solution :
(A) Tetany → (iv) Rapid spasm in muscle due to low calcium
(B) Muscular dystrophy → (iii) Degeneration of skeletal muscles, genetic origin
(C) Arthritis → (ii) Inflammation of joints
(D) Myasthenia gravis → (i) Paralysis of skeletal muscle due to autoimmune disorder

Final Answer:
Option (2)

126)

Explain Question:
Rapid muscle spasm (tetany) occurs due to imbalance of which certain ions in body fluids.

Concept:
Muscle disorder

Solution:
Calcium ions (Ca^{2+}) Regulate muscle contraction and relaxation.
Low concentration of Ca^{2+} in body fluids increases neuromuscular excitability, leading to rapid muscle spasms (tetany).

Final answer: Option (3)

127)

NCERT Pg. # 222-223

128)

Question explanation: Identify how many of the listed bones are **unpaired** in the human skeleton.
Concept: Skeletal system
Solution: The correct answer is **option 1**.

A. **Hyoid** → Unpaired

B. **Maxilla** → Paired

C. **Temporal** → Paired

D. **Mandible** → Unpaired

E. **Vomer** → Unpaired

F. **Sternum** → Unpaired

G. **Zygomatic** → Paired

Unpaired bones: Hyoid, Mandible, Vomer, Sternum → **4 bones**

Final answer: Option 1

129)

Question explanation: Match each type of joint with its correct anatomical location.
Concept: Classification of joints
Solution:

A. **Fibrous joint** → found **between cranial bones → A–(iii)**

B. **Cartilaginous joint** → found **between two adjoining vertebrae → B–(iv)**

C. **Pivot joint** → found **between atlas and axis → C–(v)**

D. **Hinge joint** → found **between femur and tibia (knee)** and **between humerus and radioulna (elbow) → D–(vi), (vii)**

E. **Gliding joint** → found **between carpals** and **between tarsals → E–(i), (ii)**

Correct matching: A–(iii), B–(iv), C–(v), D–(vi, vii), E–(i, ii)

Final answer: Option 3

130)

NCERT Pg. # 227

131)

NCERT-XI, Pg. # 28

132)

NCERT-XI, Pg. # 31, 32

133)

NCERT-XI, Pg. # 30

134)

NCERT-XI, Pg. # 33

135)

NCERT-XI, Pg. # 33

136)

NCERT-XI, Pg. # 34

137)

NCERT-XI, Pg. # 28, 29, 30, 33

138)

NCERT-XI, Pg. # 26, 27

139)

NCERT-XI, Pg. # 28

140) NCERT-XI, Pg. # 29

141) NCERT-XI, Pg. # 28

142) NCERT XI Pg. # 29

143)

NCERT-XI, Pg. # 29, 30, 32

144)

NCERT-XI, Pg. # 29

145) NCERT-XI, Pg. # 29

146)

NCERT-XI, Pg. # 30

147) NCERT XI, Pg # 30

148)

NCERT-XI, Pg. # 28

149)

NCERT-XI, Pg. # 28

150)

NCERT-XI, Pg. # 30

151) NCERT-XI, Pg. # 34

152) NCERT-XI, Pg. # 29

153) NCERT-XI, Pg. # 29

154)

NCERT-XI, Pg. # 32

155)

NCERT-XI, Pg. # 32

156)

NCERT-XI, Pg. # 31, 32, 33

157)

NCERT-XI, Pg. # 32, 33

158)

NCERT-XI, Pg. # 33

159)

NCERT-XI, Pg. # 33

160)

NCERT-XI, Pg. # 32

161)

NCERT-XI, Pg. # 33

162)

NCERT-XI, Pg. # 32

163)

NCERT-XI, Pg. # 32

164)

NCERT-XI, Pg. # 31

165)

NCERT-XI, Pg. # 32

166)

NCERT-XI, Pg. # 32

167)

NCERT-XI, Pg. # 32

168)

NCERT-XI, Pg. # 31

169) NCERT-XI, Pg. # 31, 32

170)

NCERT-XI, Pg. # 33

171)

NCERT-XI, Pg. # 31

172)

NCERT-XI, Pg. # 32

173)

NCERT-XI, Pg. # 32, 33

174)

NCERT-XI, Pg. # 31

175)

NCERT-XI, Pg. # 31, 32

176)

NCERT-XI, Pg. # 130, 131

177)

NCERT-XI, Pg. # 131

178) NCERT-XI, Pg. # 130, 131

179) NCERT-XI, Pg. # 130, 131

180)